

Exp: 6 Develop a lexical Analyzer to test whether a given identifier is valid or not using C.

```
#include <stdio.h>

#include <ctype.h>

#include <string.h>

int main() {
    char a[10];

    int flag = 1;

    int i = 1;

    printf("\nEnter an identifier: ");

    fgets(a, sizeof(a), stdin);

    size_t len = strlen(a);

    if (len > 0 && a[len - 1] == '\n') {
        a[len - 1] = '\0';
    }

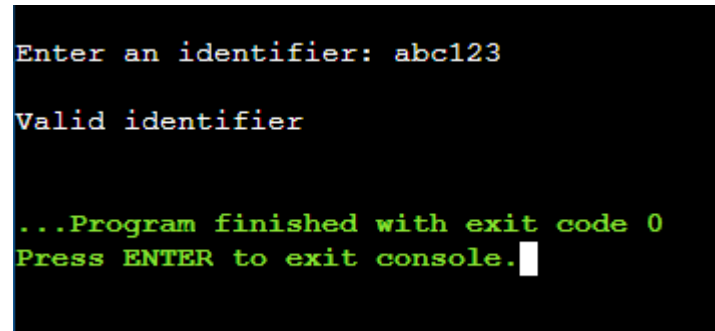
    if (isalpha(a[0]) || a[0] == '_') {
        while (a[i] != '\0') {
            if (!isdigit(a[i]) && !isalpha(a[i]) && a[i] != '_') {
                flag = 0;
                break;
            }

            i++;
        }
    } else {
        flag = 0;
    }

    if (flag == 1) {
        printf("\nValid identifier\n");
    } else {
        printf("\nNot a valid identifier\n");
    }
}
```

```
    return 0;  
}
```

Output:

A screenshot of a console window with a black background and white and green text. The text shows the program's execution flow: a prompt to enter an identifier, the user input 'abc123', a confirmation that it is a valid identifier, and a final message indicating the program finished successfully with exit code 0, prompting the user to press ENTER to exit the console.

```
Enter an identifier: abc123  
  
Valid identifier  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```